

Controlling Microbial Growth

- Microbes are ubiquitous and many are associated with undesirable consequences such as food spoilage and disease.
- it is essential to kill a wide variety of microorganisms or inhibit their growth to minimize their destructive effects
- The goal
 - to destroy pathogens and prevent their transmission
 - reduce or eliminate microorganisms responsible for the contamination of water, food, and other substances

Definition of Frequently Used Terms

- ***Sterilization***: Sterilization is the total destruction of all microbes
 - including **spores**, mycobacteria, nonenveloped (nonlipid) viruses, and fungi
- A **germicide**, also called a **microbicide**, is any chemical agent that kills pathogenic microorganisms.
- **Disinfection**, is the process of removing microorganisms, including potentially pathogenic ones, from the surfaces of **inanimate objects**
 - destroys vegetative pathogens but not bacterial endospores
 - used only on inanimate objects
 - remove the harmful products of microorganisms (toxins)
- **Decontamination** a process of treatment that renders a medical device, instrument, or environmental surface **safe to handle** but does not necessarily mean that the item is safe for patient reuse

- **Decontamination** (Cleaning, disinfection, and sterilisation) are the **backbone** of infection prevention and control
- The type and level of device **decontamination** depends upon the nature of the item and its intended use
- **E.H. Spaulding** Classification of medical/surgical devices based on their potential to spread infections
 - Critical items require **sterilization**
 - Semi critical item require **high-level disinfection**
 - Non critical items requires **a low to intermediate** level disinfectant

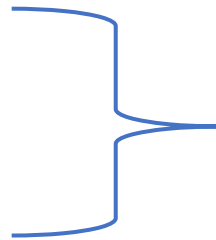
Methods of sterilization can be broadly classified as:

- Physical methods of sterilization

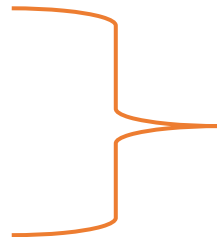
- Sunlight
- Heat
- Filtration
- Radiation
- Sound (sonic) waves

- Chemical methods of sterilization

- alcohols,
- aldehydes
- Phenols
- Organic solvents
- Heavy metals
- Oxidizing agents
- Dyes
- Alkylating agents



damage the cell wall/membrane and
denature proteins



modify functional groups of proteins and
nucleic acids

Physical means of sterilization and disinfection

🍀 Dry heat (hot air)

- Red Heat
- Flaming
- Incineration
- Infrared radiation
- Hot air oven

🍀 Moist Heat Sterilization

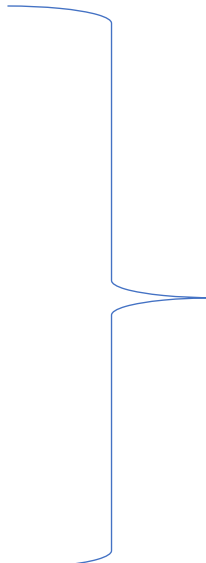
- At temperatures below 100°C
- At a temperature of 100°C
- At temperatures above 100°C

🍀 Filtration

- Filtration sterilization of liquids
- Filtration sterilization of gases

🍀 Irradiation

- Ultraviolet (non-ionizing) radiation
- Ionizing Radiation



Heat sterilization

Heat Sterilization : is the most reliable and universally applicable method of sterilization.

The process is more effective in hydrated state

Mechanism of Action:

involves destruction of enzymes and other essential cell constituents

Dry heat-denatures protein.

Moist heat-denatures and coagulates protein.

applied only to the thermostable products, but it can be used for

Moisture-sensitive materials for which **dry heat** (160-180°C) sterilization,

Moisture-resistant materials for which **moist heat** (121-134°C) sterilization is used.

The efficiency with which heat is able to inactivate microorganisms is dependent

upon the degree of heat,

the exposure time

presence of water.

Dry heat: process of removing microorganisms by applying **moisture-free heat** which is appropriate for **moisture-sensitive** substances.

More convenient for heat-stable, non-aqueous materials that cannot be sterilized by steam because of its deleterious effects or failure to penetrate

glassware, powders, oils, and some oil-based injectable

Preparations to be sterilized by dry heat are filled in units that are either sealed or temporarily closed for sterilization.

Temperature (°C)

Minimum sterilization time (min)

160

120

170

60

180

30

Moist Heat Sterilization

- Is one of the most effective methods where the **steam under pressure** acts as a bactericidal agent.
- High pressure increases the **boiling point** of water and thus helps achieve a higher temperature for sterilization.
- High pressure also facilitates the **rapid penetration** of heat into deeper parts of material and moisture present in the steam causes the **coagulation** of **proteins** causing an **irreversible loss of function** and activity of microbes.
- Moist heat sterilization involves the use of steam in the range of 121-134°C.
- An **autoclave** is a device that works on the principle of moist heat sterilization
- Basically moist heat sterilization is of 3 types
 - Temp below 100°C Pasteurization
 - Temp at 100°C
 - Temp above 100°C (Autoclave)

Sterilization at temperatures below 100°C

- The sterilization technique employed at a temperature below 100°C involves **pasteurization**.
- **Pasteurization**: application of heat at temperature of 62°C for 30 minutes or 72°C for 15 minutes followed by **rapid cooling**
- In this process, all non-spore forming microbes are killed in **milk**, fruit juices, beer, and wine by subjecting
 - T ° of 63°C for 30 min (the holder method)
 - 73°C for 20 seconds (the flash method).
- however, not all the pathogenic organisms are killed
- The principle of pasteurization is the **logarithmic reduction** in the number of **viable** microbes so that they can no longer cause diseases.
- All **mesophilic non-sporing** bacteria can be killed by exposure to a moist heat at 60°C for half an hour
- Besides milk, other fluids and equipment like vaccines of non-sporing bacteria are also pasteurized at 60°C for 1 hour in special water baths.

Sterilization At a temperature of 100°C

- Boiling at 100°C **doesn't ensure complete sterility**, but is enough for the removal of **pathogenic vegetative** microbes and **some spores**.
- **Tyndallisation** (Intermittent steaming) Steaming of the material is done at 100 °C for 30 minutes on three consecutive days.
 - The **principle** is that spores which survived the heating process would germinate before the next thermal exposure and then would be killed.
 - It is used for sterilizing **heat sensitive** culture media containing materials such as **carbohydrate, egg, gelatin** or **serum**.
 - Is also applicable for contaminated dishes, beddings, pipettes, and other instruments that are not soiled or contaminated as well as for objects that are temperature sensitive.

At temperatures above 100°C (Autoclave)

Most effective method of decontamination

During autoclaving, pressure is used to produce high temperature steam to achieve sterilization

all bacteria, bacterial spores, viruses, fungi, and protozoa

Hot saturated steam in autoclaving acts as an **excellent agent** for sterilization because of:

- High temperature;
- Wealth of latent heat (steam under pressure);
- It destroys bacterial spores and vegetative cells.

Sterilizing reusable syringes, specimen containers, tubes, pipettes, petri dishes, and other

Time-Temperature-Pressure Level Relationship in moist heat sterilization (Autoclaving)

Temperature	Time	Pressure level
121 ^o c	15 minutes	15 lb/inch ²
126 ^o c	10 minutes	20 lb/inch ²
134 ^o c	3 minutes	30 lb/inch ²



Filtration

- Is unique among sterilization techniques in that it **removes**, rather than destroys, microorganisms.
- Uses **membranous filters** that have tiny pores can **prevent bigger particles** such as bacteria from passing through the filter.
- capable of preventing the passage of both viable and nonviable particles and can thus be used for both the **clarification** and **sterilization** of liquids and gases.
- involves **sieving**, **adsorption**, and **trapping** within the matrix of the filter material.
- Certain types of filter (membrane filters) also have an essential role in **sterility testing**
- The principal application of sterilizing-grade filters is the **treatment** of **heat-sensitive injections** and **ophthalmic solutions**, biological products, air, and other gases for supply to aseptic areas.

Filtration

- **Liquids** and **gases** can be sterilized by passing them through filters. Filters that have pores smaller than the size of the microbes retain micro-organisms.
- Standard bacteriological membrane filters are composed of nitrocellulose and have pore diameters of **0.45 μm** , small enough to prevent passage of most bacteria.

Uses :

- Preparing heat-labile culture media components
- Preparing pharmaceuticals and biological solutions
- Microbial evaluation of water purity
- Bacterial viability count
- Determination of viral particle size

Irradiation: is the process of exposing surfaces and objects to different kinds of radiation for sterilization

Mainly **electromagnetic** radiation is used for sterilization.

The major target is **microbial DNA**, where damage occurs as a result of ionization and free radical production (gamma-rays and electrons) or excitation (UV light).

Gamma rays, x-rays, beta rays, cosmic rays, ultraviolet light, and even visible light are all forms of radiation. When these rays strike an organism, energy may be absorbed by the cells, often causing cell damage or death.

▪ Are divided into **ionic** and **ultra violet** radiation

Ionizing Radiation

- **X-ray** and **gamma** rays are the commonly used ionizing radiation for sterilization.
- Penetrate deep into objects and destroy bacterial endospores and vegetative cells
 - not effective against viruses
- The ionization results in the formation of a large number of **toxic O₂ metabolites** like hydroxyl radical, superoxide ion, and H₂O₂ through ionization of water.
- These metabolites are highly oxidizing agents and kill microorganisms by oxidizing various cellular components.
- Radiation sterilization is generally applied to items in the dried state which include **surgical instruments, sutures, prostheses, unit-dose ointments, plastic syringes, and dry pharmaceutical** products.

Ultraviolet (non-ionizing) Radiation

- Ultraviolet radiation includes light rays from 150-3900 Å, of which 2600 Å has the highest bactericidal effect.
- It has less quantum energy with **low penetrating** than ionic radiation so microorganisms only on the surface are killed.

Upon exposure, these waves are absorbed by many materials, particularly nucleic acids.

Cause the formation of **pyrimidine dimers** which bring error in DNA replication and cause the death of microbes by mutation.

UV radiation owing to its poor penetrability of conventional packaging materials is **unsuitable** for sterilization of **pharmaceutical dosage** forms.

It is, however, applied in the sterilization of **air**, for the **surface sterilization** of aseptic work areas, and the treatment of manufacturing-grade water.

Activities of commonly used disinfectants

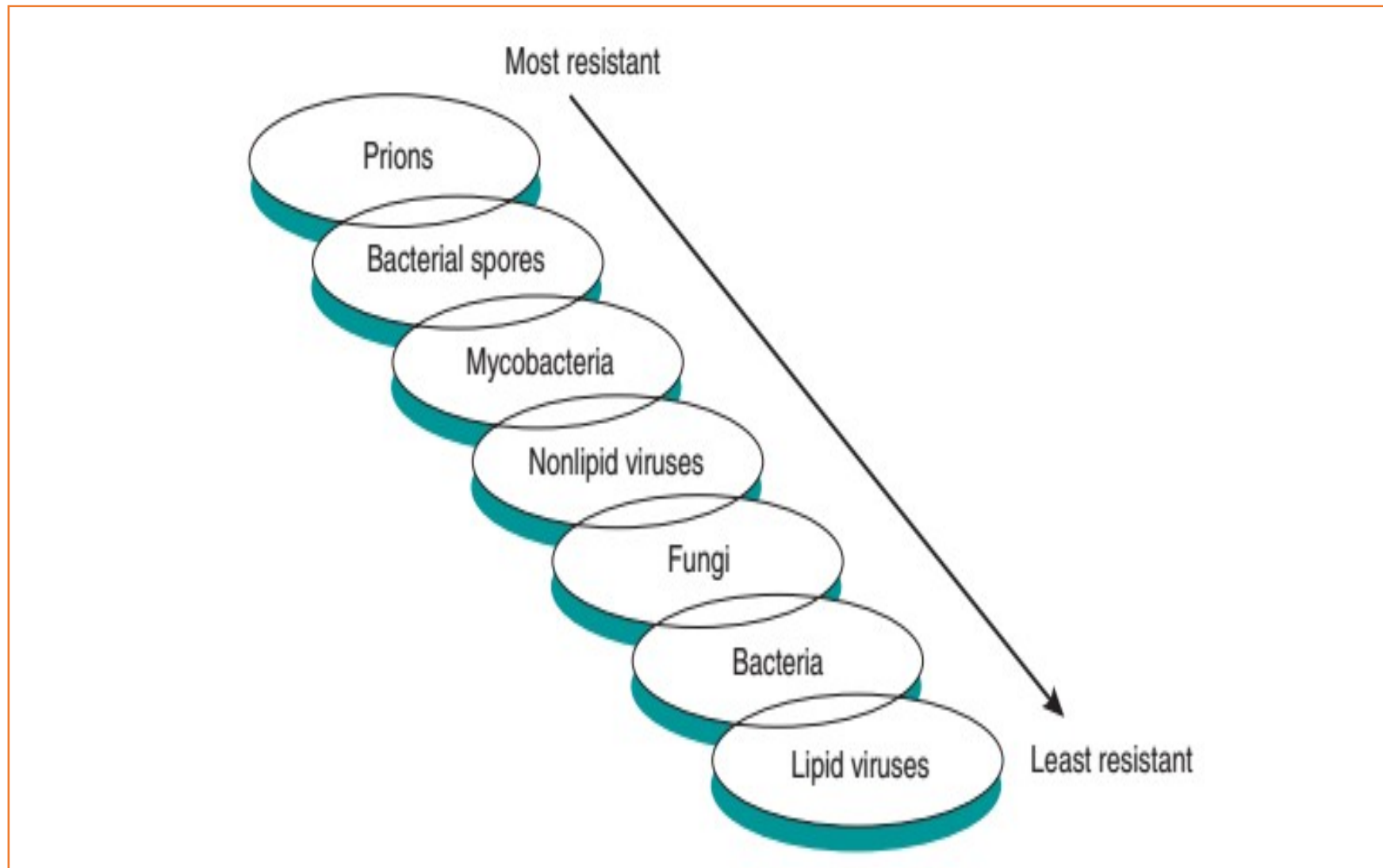
Disinfectant	Fungi	Bacteria	Spores	Enveloped viruses	Nonenveloped viruses
Phenol	Good	Good	Nil	Good	Variable
Hypochlorite	Good	Good	Fair	Good	Good
Alcohols	Good	Good	Nil	Good	Variable
Aldehydes	Good	Good	Good	Good	Good
Glutaraldehyde	Good	Good	Good	Good	Good
Iodophor	Good	Good	Nil	Good	Good

Properties of Ideal Disinfectant

- wide spectrum of antimicrobial activity (Gram-positive and Gram-negative bacteria, acid-fast bacteria, bacterial endospores, fungi, and viruses)
- It should act in the presence of organic matter.
- none toxic to human or corrosive
- stable upon storage and should not undergo any chemical change
- soluble in water and lipids for penetration into microorganisms
- effective in acidic as well as in alkaline media

Factors Influencing Activity of Disinfectants

- Various conditions influencing the efficiency of disinfectant are as follows:
 - Temperature: Increase in temperature increases the efficiency
 - Type of microorganism: Spores may be resistant to the action of disinfectants.
 - Physiological state of the cell: Young and metabolically active cells are more sensitive
 - Environment: physical or chemical properties of the medium
 - Number of organisms(microbial load)
 - Concentration of disinfecting agent
 - Contact time
 - pH
 - Biofilms
 - Compatibility of disinfectants and sterilants



Different types of organisms and their resistance to killing agents.